

Course: Data Structures (CSE CS203A, 114-1)
Quiz II: Array, Linked List, Stack and Queue
October 21, 2025, 16:30~17:00

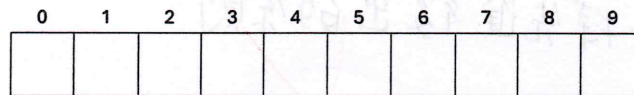
55 + 10

Student ID: 1123301

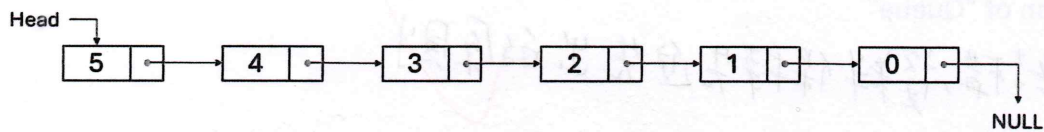
Student Name: 黃立達

Data Structures: Visualization

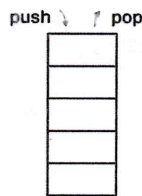
(1) Array



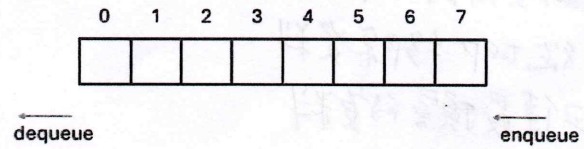
(2) Linked List



(3) Stack



(4) Queue



Q1: (30 pts; 10 pts for each) Describe the mechanism of the function

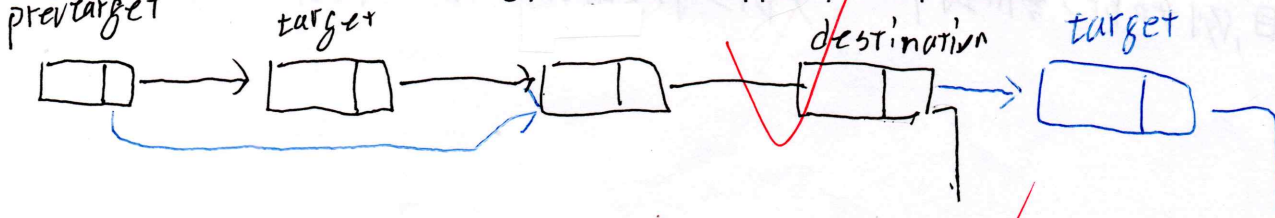
MoveTo(node *head, node *target, node*destination)

A1: Write a short paragraph explaining how the **MoveTo** function works (you may answer in English or Mandarin).

① Are there any **additional variables** required? If so, explain why they are necessary.

Yes, 因為這 list 是單向的, 所以必須從 head 開始, 並用額外變數記找到 target 的 prev 位置。
打做 moveTo, 因為 prev 接到下一個 target 的

② Draw a visualization of the singly linked list to support your explanation.



③ Is there any **variation of a linked list** (e.g., doubly linked list or circular linked list) that can simplify or improve this operation?

Yes, double link list 可以避免從頭開始找 target 的 prev, 因為雙向會直接記 target prev 的位置

Q2: (40 pts, 10 pts for each) **Definition of Data Structures**

Define the following data structures and list their fundamental operations.

A2:

① Definition of "Stack"

一個資料結構, 資料保持先進後出的原則

② Definition of "Queue"

一個資料結構, 資料保持先進先出的原則

③ Preliminary operations of "Stack"

push, 放資料到 top

pop, 從 top 移除資料

top, 回傳最上層的資料

④ Preliminary operations of "Queues"

push, 放資料到最後

pop, 從前面移除資料

front, 回傳最前面的資料

Q3: (30 pts) **AI Copilot Application**

Choose up to two data structures from the visualization list above.

Compose a single prompt (within 300 words) that you would use with an AI Copilot to explore or learn advanced concepts related to your chosen data structures.

A3:

我會先問架構, 然後讓他跑用案例的方式實際讓我看到資料跟架構如何運作, 然後開始問詳細的問題, 像是在某某情況下那一刻能較好, 還有像是空間分配與不同情況的實做難度等問題, 然後我還會問會用到此結構的題目, 例如 BFS 會用到 queue, DFS 可以用 stack 實作等。

—30